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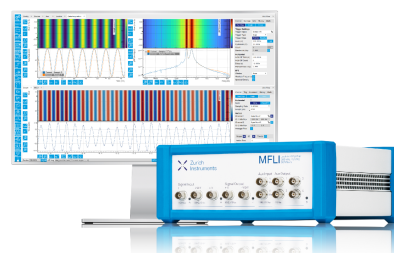
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Angular scattering of 1–50 keV ions through graphene and thin carbon foils: Potential applications for space plasma instrumentation

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We present experimental results for the angular scattering of ~ 1 –50 keV H, He, C, O, N, Ne, and Ar ions transiting through graphene foils and compare them with scattering through nominal $\sim 0.5 \mu\text{g cm}^{-2}$ carbon foils. Thin carbon foils play a critical role in time-of-flight ion mass spectrometers and energetic neutral atom sensors in space. These instruments take advantage of the charge exchange and secondary electron emission produced as ions or neutral atoms transit these foils. This interaction also produces angular scattering and energy straggling for the incident ion or neutral atom that acts to decrease the performance of a given instrument. Our results show that the angular scattering of ions through graphene is less pronounced than through the state-of-the-art $0.5 \mu\text{g cm}^{-2}$ carbon foils used in space-based particle detectors. At energies less than 50 keV, the scattering angle half width at half maximum, $\psi_{1/2}$, for ~ 3 –5 atoms thick graphene is up to a factor of 3.5 smaller than for $0.5 \mu\text{g cm}^{-2}$ (~ 20 atoms thick) carbon foils. Thus, graphene foils have the potential to improve the performance of space-based plasma instruments for energies below ~ 50 keV. © 2014 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4866850>]

I. INTRODUCTION

Thin carbon foils have become a critical component for space-based time-of-flight (TOF) ion mass spectrometers and energetic neutral atom (ENA) sensors.¹ These instruments take advantage of the charge exchange and secondary electron emission produced as incident ions and neutral atoms transit thin carbon foils. This interaction also creates several adverse effects for the projectile exiting the foil. For example, angular scattering and energy straggling act to limit the performance of a given instrument.^{1,2} The magnitude of these effects varies with the incident angle, energy, and mass of the incoming projectile and the foil thickness; minimizing the foil thickness is imperative for achieving optimal instrument performance. One approach is to fly very thin foils, the current state-of-the-art being carbon foils with a nominal thickness (as specified by the manufacturer) of $\sim 0.5 \mu\text{g cm}^{-2}$. Using the measured density of $\sim 1.8 \text{ g cm}^{-3}$ for thin carbon foils,³ this corresponds to a thickness of $\sim 3 \text{ nm}$ or ~ 20 atoms. The actual measured foil thickness is generally larger than its stated nominal value due to the presence of an additional $\geq 0.5 \mu\text{g cm}^{-2}$ impurity layer on the foil surface that is thought to be composed partly of residual parting agent used in producing the foil and absorbed gases.⁴ This increased thickness has the effect of further increasing the angular scattering and energy loss for the incident particle.

Recently, a new type of thin carbon surface has emerged – graphene.^{5,6} Graphene is composed of carbon atoms that are

tightly packed into a two-dimensional (2D) honeycomb lattice a single atom thick. It can be rolled (fullerenes), wrapped (nanotubes), and layered (graphite) to produce different types of carbon-based materials.^{7,8} Since first being isolated in 2004 via micromechanical cleavage of graphite, graphene's electronic and structural properties have been a topic of great interest to the physics community, the 2010 Noble Prize in physics being awarded to the group first credited with measuring its properties.⁹ From a space science perspective, graphene's thin nature and considerable strength (Young modulus $\sim 1 \text{ TPa}$)¹⁰ and elasticity (spring constant ~ 1 –5 N m^{-2})¹¹ make it a potential foil candidate that may be able to further decrease some of the undesirable effects associated with thin carbon foils and improve the performance of plasma instruments in space.

In this paper, we present experimental results for the angular scattering of ~ 1 –50 keV H, He, C, O, N, Ne, and Ar ions transiting through graphene and compare them with results for state-of-the-art carbon foils collected under similar experimental conditions. To our knowledge, this is the first study to quantify the angular scattering of 1–50 keV ions through graphene and represents one of a series of studies¹² that examine the potential of graphene as a foil candidate for space-based particle sensors.

II. EXPERIMENT AND ANALYSIS

All testing was performed in the Ion Calibration Facility at Southwest Research Institute (SwRI). This ultrahigh vacuum (UHV) facility uses a duo-plasmatron source to produce

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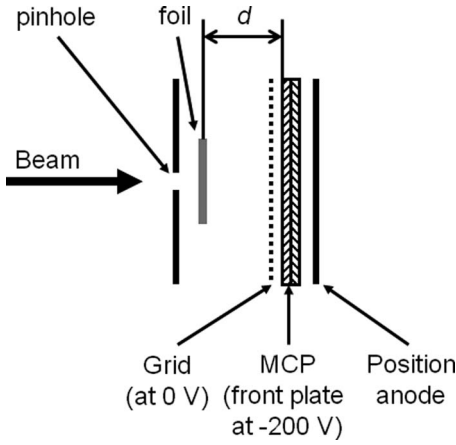


FIG. 1. Schematic of the experimental setup used in this study. A fraction of the incident ion beam passes through a ~ 1 mm diameter pinhole and then through the foil. The scattered particles are detected by a position sensitive anode located on the backside of a microchannel plate (MCP).

a focused ion beam with energies ranging from ~ 0.5 –51 kilo-electron volts per charge (keV/Q) for a mass per charge range of 1 to >40 amu/Q. The ions are mass per charge selected using a Wien filter that provides a mass resolution ($M/\Delta M$) of ~ 50 . A detailed description of this facility can be found in Ref. 13.

Figure 1 shows a schematic of the experimental setup used in this study. A fraction of the ion beam incident on the experiment passes through an ~ 1 mm diameter pinhole before interacting with graphene or a thin carbon foil. The foils were affixed to 333 or 1000 lines/in. (lpi) nickel grids and mounted on a wheel capable of holding up to 15 different samples. The wheel is actuated with a motor, allowing for testing of many foils under similar conditions without having to break the $\sim 10^{-7}$ Torr ($\sim 10^{-5}$ Pa) vacuum. The scattered particles emitted from the backside of the foil are detected by a position sensitive microchannel plate (MCP) located a distance $d = 88$ mm from the foil. The front plate of the MCP stack is biased to -200 V in order to repel secondary electrons exiting from the foil. We collected measurements for various beam energies and species and repeated the measurements for foils of different type and thickness. The angular scattering distribution of the ions can be readily determined from this setup.

Figure 2 shows a typical image for a distribution of scattered particles exiting the backside of a foil from our setup (bottom left). The colors correspond to the number of particle counts per pixel, green being more intense than blue. Projections of the counts on the x- and y-axis are shown in black on the top and the right side of the figure, respectively. The cyan curves are mirror images of the black curves and are used to determine the location of the peak in the angular scattering distribution as indicated by the straight black line. The peak in the distribution is the location where zero scattering occurs. We construct the angular scattering distribution by first calculating the scattering angle, θ , for each of the image pixels using its distance from the peak and d . We then calculate $N(\theta, \Delta\theta)$ where N is the number of particles that have a scattering angle between $\theta - \Delta\theta/2$ and $\theta + \Delta\theta/2$. Finally, we

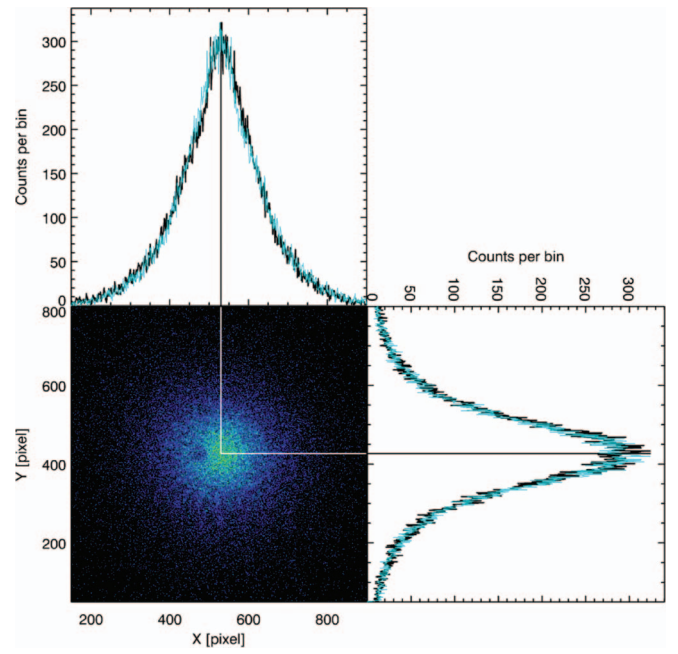


FIG. 2. Example of an angular scattering distribution for 4 keV oxygen ions passing through a $0.5 \mu\text{g cm}^{-2}$ carbon foil. The location of the distribution peak is found using the x- and y-axis projections. See text for more details.

normalize $N(\theta, \Delta\theta)$ by the solid angle $\Delta\Omega$ defined as the shell of a cone of apex angle θ and width $\Delta\theta$ as given by

$$\Delta\Omega = 2\pi \left[\left(1 - \cos \left(\theta + \frac{\Delta\theta}{2} \right) \right) - \left(1 - \cos \left(\theta - \frac{\Delta\theta}{2} \right) \right) \right] = 4\pi \sin \theta \sin \frac{\Delta\theta}{2}. \quad (1)$$

Figure 3 shows the normalized angular scattering distribution versus scattering angle θ for the example in Figure 2. The bin size is 0.5° and the data are normalized to the number of counts in the first bin ($\theta = 0.25^\circ$). The red curve is a Kappa distribution fit to the data using the following equation,

$$F(\theta) = F_o \left(1 + \frac{\theta^2}{\sigma^2 \kappa} \right)^{-\kappa}, \quad (2)$$

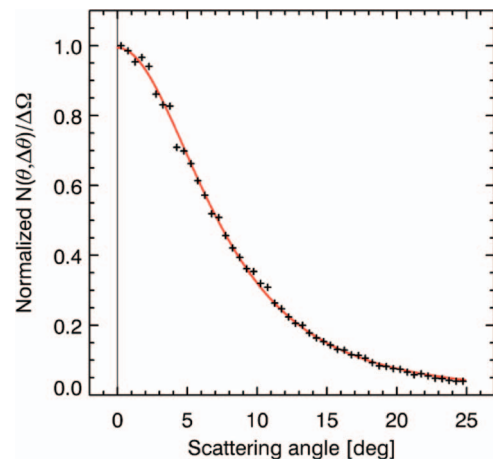


FIG. 3. Normalized angular scattering distribution versus scattering angle (black crosses) from the example in Figure 2. Red curve: Kappa distribution fit to the data.

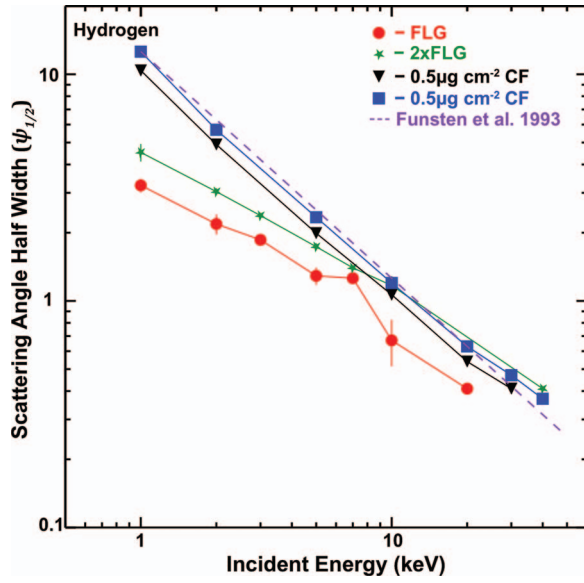


FIG. 4. Scattering angle half width at half maximum, $\psi_{1/2}$, versus incident energy for 1–50 keV hydrogen ions exiting the backside of 3–5 atomic layer thick graphene (FLG), graphene of twice that thickness (2xFLG), and two carbon foils (CFs) with a nominal thickness of $\sim 0.5 \mu\text{g cm}^{-2}$. The CF results are in good agreement with previous studies.¹⁴ The error bars reflect the standard deviation for cases with multiple measurements.

where F_0 , σ , and κ are the free parameters. The scattering angle half width at half maximum, $\psi_{1/2}$, of the distribution is given by $\psi_{1/2} = \sigma \sqrt{\kappa(\sqrt{2} - 1)}$ and is the parameter that we focus on in this study.

III. RESULTS

We present the scattering angle half width at half maximum, $\psi_{1/2}$, for the particle distributions produced by 1–50 keV H, He, C, N, O, Ne, and Ar ions exiting the backside of graphene and thin carbon foils. The graphene used in this study was ~ 3 –5 atomic layers thick (few layer graphene – FLG) or twice that thickness (2xFLG), while the carbon foils (CFs) had a nominal thickness of $0.5 \mu\text{g cm}^{-2}$ (~ 20 atomic layers), the thinnest foils typically flown in space-based plasma and ENA sensors.

Figure 4 shows $\psi_{1/2}$ versus energy for incident 1–50 keV hydrogen ions after passing through each foil. The values of $\psi_{1/2}$ were corrected to remove the contribution from the ~ 1 mm diameter beam which was only significant for cases where $\psi_{1/2}$ was $< 1^\circ$ (e.g., $\sim 10\%$ at $\psi_{1/2} = 0.5^\circ$). The half-width of the angular scattering distributions show a strong dependence on incident ion energy, consistent with previous results.¹⁴ Our CF results are also consistent with previous studies,¹⁴ thereby validating our method. The scattering angle half width for the FLG foil was narrowest throughout the energy range studied here. The largest difference between the graphene and carbon foil results is at 1 keV where the FLG foil has the narrowest scattering angle half-width ($\psi_{1/2} = 3.2^\circ$) followed by the 2xFLG foil ($\psi_{1/2} = 4.5^\circ$) and then the CFs ($\psi_{1/2} = 10.4^\circ$ – black symbol; $\psi_{1/2} = 12.6^\circ$ – blue symbol). $\psi_{1/2}$ for the 2xFLG foil is narrower than the CFs up to ~ 10 keV and is similar above that energy.

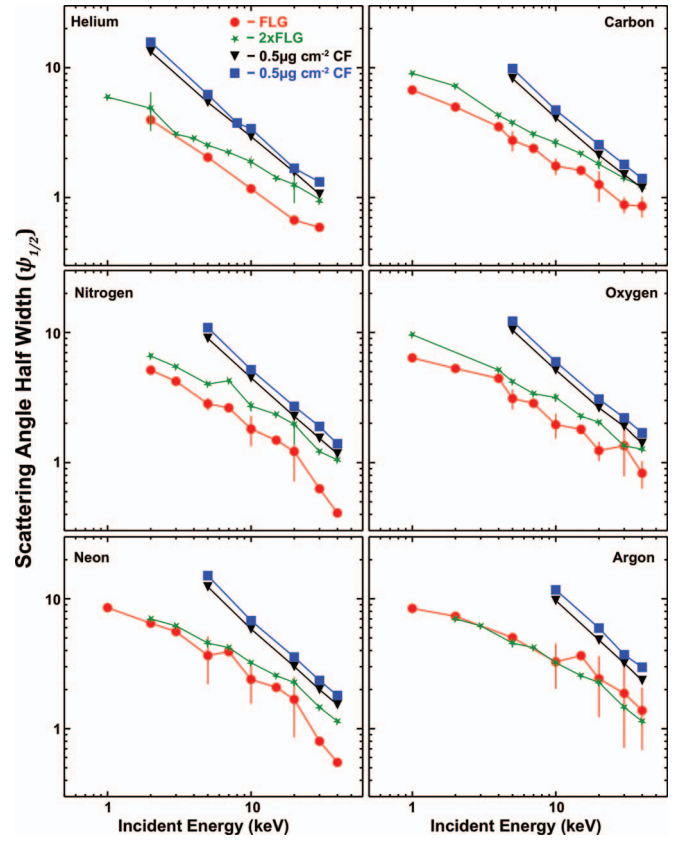


FIG. 5. [(a)–(f)] Same format as Figure 4 except for He, C, O, N, Ne, and Ar ions.

Another interesting result is that the energy-dependence of $\psi_{1/2}$ for hydrogen ions passing through graphene is different than for the thin carbon foils. The measured values of $\psi_{1/2}$ for the CFs are inversely proportional the incident energy, consistent with the Meyer theory for multiple scattering of low-energy particles in solids.¹⁵ $\psi_{1/2}$ for graphene has a flatter slope in this representation, especially below ~ 10 keV. Our interpretation of these results will be discussed in Sec. IV.

Figures 5(a)–5(f) show plots of $\psi_{1/2}$ versus incident energy for the heavier than hydrogen ions studied here. $\psi_{1/2}$ for graphene is narrower than the CFs for nearly all the species and energies shown here. The difference in $\psi_{1/2}$ between the two foil types at the lowest energies reported here – 2 keV for He, 5 keV for C, N, O, Ne, and 10 keV for Ar – ranged from a factor ~ 3 –3.4 between the FLG foil and CF and ~ 2.2 –2.7 between the 2xFLG foil and CF. $\psi_{1/2}$ for graphene and the CFs also showed a strong mass dependence with $\psi_{1/2}$ increasing with mass for ions with similar energies. And similarly to hydrogen, the slope for graphene is not as steep as it is for the carbon foils.

IV. DISCUSSION AND CONCLUSIONS

Thin carbon foils have become a critical subsystem for TOF ion mass spectrometers and ENA sensors in space. Scattering of the incident ion or neutral atom as it exits the foil surface can result in particle loss (the particle does not reach the

detector because it is scattered too much) or deterioration of the measurements by introducing an uncertainty in the flight path length. The result is a decrease in instrument performance. For ENA sensors, thin carbon foils are used for charge conversion – converting incident neutral atoms into ions – and secondary electron emission.¹⁶ Similarly to the TOF measurement, scattering from the foils can result in a loss of signal to the detector thereby reducing the overall performance of the instrument.

Our results show that the angular scattering of 1–40 amu ions through graphene is less pronounced than through the state-of-the-art carbon foils used in space-based particle detectors for energies between ~ 1 and 50 keV. At the low end of this energy range, the scattering angle half width at half maximum, $\psi_{1/2}$, for 3–5 atoms thick graphene is up to a factor of ~ 3 –3.5 smaller than for $0.5 \mu\text{g cm}^{-2}$ (~ 20 atoms thick) carbon foils and is mass and energy dependent. Thus, substituting thin carbon foils with graphene has the potential to improve the performance of particle instruments for energies below ~ 50 keV.

Another interesting result is that the energy-dependence of $\psi_{1/2}$ for ions passing through graphene is different than for the carbon foils. The scattering angle half-widths for the carbon foils are inversely proportional the incident energy, consistent with the Meyer theory¹⁵ for multiple scattering (>20) of low-energy particles through a thin carbon surface. $\psi_{1/2}$ for graphene has a flatter slope. This scattering is driven by elastic encounters between the incident particle and electrons and atoms in the target material. The thickness of and, thus, the number of collisions, n , within the material play an important role in the resulting angular scattering distributions. One of the assumptions of Meyer's theory is that the foil thickness has to be large enough such that $n \gg 5$. For 3–5 atoms thick graphene, the number of collisions is expected to be less than 5 ($n < 5$) and the theory should not apply. It is plausible that a layer of impurity also exists at the surface of the graphene foil making it thicker than its nominal value. Still, the lower number of collisions for graphene may imply a breakdown of the multiple small scattering assumption in Meyer's theory and explain why $\psi_{1/2}$ for graphene is not inversely proportional to the energy of the incident ion.

This work, coupled with recent results from Ref. 12 that show ions transiting graphene and carbon foils to have similar charge state distributions exiting the foils, supports further investigation of graphene as a foil candidate for space-based plasma instruments. The next step could be to quantify the ion transmission and energy loss through the foil. Any improvement in these properties compared to thin carbon foils could provide at least two benefits for particle instruments in

space. Either instruments could be made smaller without loss of sensitivity or more sensitive instruments are possible with equivalent resources.

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